

Multiple Choice (10 marks)

1. What is the rank of the following matrix? $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
2. __ refers to a model that can neither model the training data nor generalize to new data.
 - (a) good fitting
 - (b) overfitting
 - (c) underfitting
 - (d) all of the above
3. Compared to the variance of the Maximum Likelihood Estimate (MLE), the variance of the Maximum A Posteriori (MAP) estimate is _____
 - (a) higher
 - (b) same
 - (c) lower
 - (d) it could be any of the above
4. Existential risks posed by AI are most commonly associated with which of the following professors?
 - (a) Nando de Freitas
 - (b) Yann LeCun
 - (c) Stuart Russell
 - (d) Jitendra Malik
5. The numerical output of a sigmoid node in a neural network:
 - (a) Is unbounded, encompassing all real numbers.
 - (b) Is unbounded, encompassing all integers.
 - (c) Is bounded between 0 and 1.
 - (d) Is bounded between -1 and 1.

True / False (10 marks)

Answer True or False

6. True or False: To calculate $P(H|E, F)$, the sufficient numbers are $P(E, F)$, $P(H)$, and $P(E, F|H)$.
7. True or False: The Bayesian network $H \rightarrow U \leftarrow P \leftarrow W$ requires only 4 independent parameters to fully specify its conditional probabilities.
8. True or False: The ResNet-50 model contains over 1 billion parameters, making it one of the largest neural networks available.
9. True or False: Ridge regression is preferred for feature selection as it can identify and eliminate less important features from the model.
10. True or False: The models obtained from best-subset selection and forward stepwise selection are always identical to each other.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether all the bits are unset in the given range or not.
12. Write a function to check if the given expression is balanced or not.

End of Paper